

flow, Puppet checks for the current status of the target machine and will only make changes when there is any specific change in the configuration. Idempotency helps in managing any particular machine throughout its life cycle, starting from the creation of the machine and the configurational changes in it, till its end-of-life.

- It supports full scale automation with proper reporting and compliance features.

Official website: <https://puppet.com>

Latest version: 6.0.1

Fuel

Fuel is an open source, GUI based tool for systems administrators to deploy and manage OpenStack Cloud along with various components. It accelerates the complex, time-consuming, error-prone task of deploying and running various configuration flavours of OpenStack. Fuel automatically discovers all virtual nodes connected from the network. It provides a complete picture regarding the nodes ready for allocation, enabling systems administrators to assign roles and resources across the cloud.

Fuel provides a step-by-step wizard for systems administrators to select the following: the host OS, hypervisors, storage back-ends, network topology, controller configuration specifications and controllers deployed on nodes.

Architecture

Figure 6 illustrates the architecture of Fuel.

Components

The following components make up Fuel.

- **Web UI:** This provides an interface to administrators to help them use the basic capabilities of Fuel, and this application is written in JavaScript.
- **Nailgun:** This implements a REST API as well as deployment data management. It manages disk volumes, configuration data, network configuration data and any other environment-specific data necessary for successful deployment. It is written in the Python language.
- **Astute:** This represents Nailgun's workers and encapsulates all details regarding the interactions with a variety of services like Cobbler, Puppet, shell scripts, etc, and provides a universal asynchronous interface to these services.
- **Cobbler:** This is used as a provisioning service.
- **Puppet:** This is a deployment service and creates MCollective agents to manage other configuration management frameworks like Chef, SaltStack, etc.
- **MCollective agents:** These perform tasks like hard drive cleaning, network connectivity probing, etc.
- **OSTF:** The OpenStack Testing Framework implements post-deployment verification of OpenStack.

Features

Listed below are the features of Fuel:

- It helps systems administrators to deploy, manage and scale multiple OpenStack clusters.
- It allows hardware configuration using the GUI.
- It performs post-deployment checks and tests for validating the OpenStack Cloud deployed.
- It helps the sysadmin to monitor real-time logs via the GUI of OpenStack Cloud.
- It performs pre-deployment checks and network validations.

Official website: <https://launchpad.net/fuel>; <https://wiki.openstack.org/wiki/Fuel>

Latest version: 11.0

Canonical OpenStack Autopilot

Canonical OpenStack Autopilot is the easiest tool for OpenStack installation. It makes use of three tools from Canonical — MAAS, Juju and Landscape. Canonical's Metal as a Service (MAAS) is a provisioning software designed to commission physical servers instantaneously.

The key components of a typical MAAS environment include a region controller, rack/cluster controller(s) and target nodes (physical servers). MAAS uses the Asynchronous Messaging Protocol (AMP) based RPC mechanism. It connects processes in the region to processes in the cluster. Juju is an open source model-driven service orchestration tool, which allows sysadmins to deploy, configure, manage and scale applications quickly and efficiently on the cloud as well as on physical servers. Landscape is a systems management tool to administer and audit large pools of Ubuntu servers. Systems are monitored by Landscape through a management agent installed on each machine. The agent automatically sends a selected set of essential health metrics, resource utilisation stats and other data to the Landscape server. Using Landscape, administrative tasks such as package and repository management, frequent upgrades, updating packages,

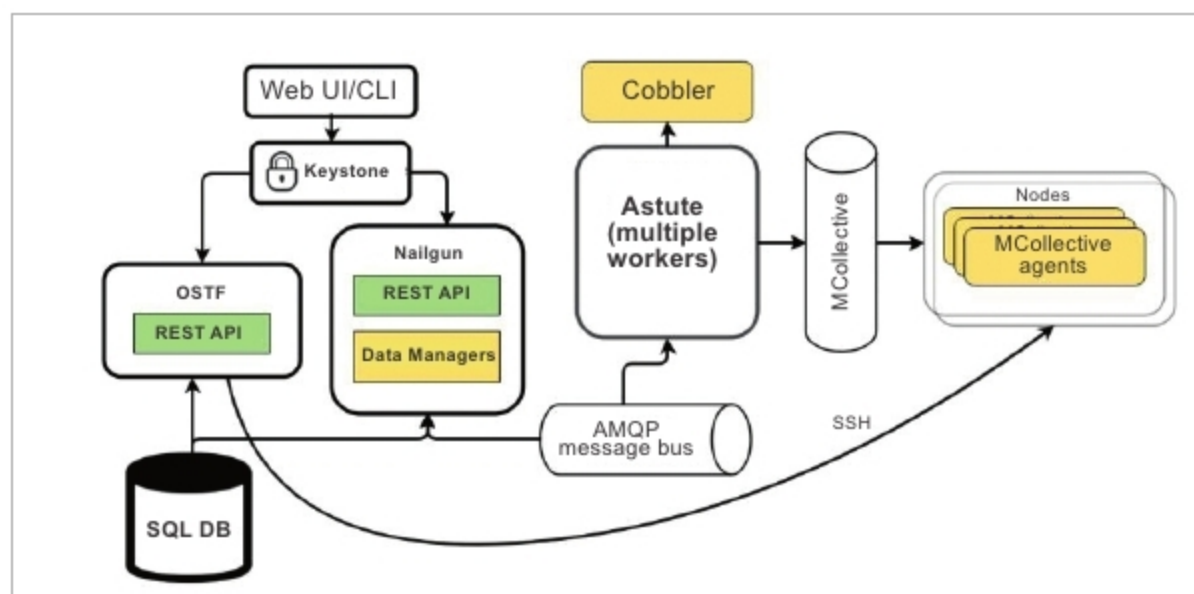


Figure 6: Fuel architecture